PEARSON EDEXCEL INTERNATIONAL A LEVEL · CHEMISTRY · WCH12

08

Redox Chemistry and Groups 1, 2 and 7

Oxidation numbers, Group trends, and the halogens that prove them.

Oxidation numbers and their rules; electron transfer and redox; disproportionation; ionic half-equations; Group 1 and 2 reactivity, oxides, solubility and thermal stability; flame colours and ion tests; titrations and uncertainty; Group 7 oxidising power, displacement, halide reactions and predictions.

UNIT	SECTIONS	STUDY TIME	FLASHCARDS
2 · WCH12	29	~7 h	17

COVERS **29 SPECIFICATION POINTS** — STATEMENTS QUOTED VERBATIM FROM THE SPECIFICATION

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8.28 Predicting Fluorine and Astatine

SPEC 8.28 · 8 min

HOW TO USE THIS DOCUMENT

Every section opens with the specification statement, worded exactly as Edexcel words it — that is what the examiner marks against. Definitions, equations and exam alerts are boxed; worked examples walk the full method step by step. Read a section, shut the page and recall its bullets from memory, then grind the topic's flashcard deck on the site.

SPEC 8.1 - 8.28 ≈ 10 MIN

Specification Map

THE SPECIFICATION SAYS

Students will be assessed on their ability to:

Every statement the specification assesses in this topic — use it as your revision checklist.

SPEC	ASSESSMENT STATEMENT
8.1	know what is meant by the term 'oxidation number' and understand the rules for assigning oxidation numbers
8.2	be able to calculate the oxidation number of elements in compounds and ions, including in peroxides and metal hydrides
8.3	be able to indicate the oxidation number of an element in a compound or an ion, using a Roman numeral
8.4	be able to write formulae given oxidation numbers
8.5	understand oxidation and reduction in terms of electron transfer and changes in oxidation number, and the application of these ideas to reactions of s-block and p-block elements
8.6	know that oxidising agents gain electrons and reducing agents lose electrons
8.7	understand that a disproportionation reaction involves an element in a single species being simultaneously oxidised and reduced
8.8	know that oxidation number is a useful concept in terms of the classification of reactions as redox and as disproportionation
8.9	understand that metals, in general, form positive ions by loss of electrons with an increase in oxidation number whereas non-metals, in general, form negative ions by gain of electrons with a decrease in oxidation number
8.10	be able to write ionic half-equations and use them to construct full ionic equations
8.11	understand reasons for the trend in ionisation energy down Groups 1 and 2
8.12	understand reasons for the trend in reactivity of the elements down Group 1 (Li to K) and Group 2 (Mg to Ba)
8.13	know the reactions of the elements of Group 1 (Li to K) and Group 2 (Mg to Ba) with oxygen, chlorine and water
8.14	know the reactions of: i oxides of Group 1 and 2 elements with water and dilute acid ii hydroxides of Group 1 and 2 elements with dilute acid
8.15	know the trends in solubility of the hydroxides and sulfates of Group 2 elements
8.16	understand the reasons for the trends in thermal stability of the nitrates and the carbonates of the elements in Groups 1 and 2 in terms of the size and charge of the cations involved
8.17	understand the formation of characteristic flame colours by Group 1 and 2 compounds in terms of electron transitions

SPEC	ASSESSMENT STATEMENT
8.18	know experimental procedures to show: i patterns in the thermal decomposition of Group 1 and 2 nitrates and carbonates ii flame colours in compounds of Group 1 and 2 elements
8.19	know reactions, including ionic equations where appropriate, for identifying: i carbonate ions, CO_3^{2-}, and hydrogencarbonate ions, HCO_3^-, using an aqueous acid to form carbon dioxide (and testing the gas with limewater) ii sulfate ions, SO_4^{2-}, using acidified barium chloride solution iii ammonium ions, NH_4^+, using sodium hydroxide solution and warming to form ammonia (and testing with litmus and HCl fumes)
8.20	be able to calculate solution concentrations, in mol dm^{-3} and g dm^{-3}, including simple acid–base titrations using the indicators methyl orange and phenolphthalein
8.21	CORE PRACTICAL 3 — finding the concentration of a solution of hydrochloric acid
8.22	understand how to minimize the sources of measurement uncertainty in volumetric analysis and estimate the overall uncertainty in the calculated result
8.23	CORE PRACTICAL 4 — preparation of a standard solution from a solid acid and use it to find the concentration of a solution of sodium hydroxide
8.24	understand reasons for the trends for Group 7 elements in: i melting and boiling temperatures and physical state at room temperature ii electronegativity iii reactivity down the group
8.25	understand the trend in reactivity of Group 7 elements in terms of the redox reactions of Cl_2, Br_2 and I_2 with halide ions in aqueous solution
8.26	understand, in terms of changes in oxidation number, the reactions of the halogens with Group 1 and 2 metals, with water, with cold dilute aqueous sodium hydroxide and with hot alkali — and reactions analogous to these
8.27	understand the reactions of: i solid Group 1 halides with concentrated sulfuric acid ii aqueous Cl^-, Br^- and I^- with silver nitrate solution and nitric acid, and the solubility of the precipitates in aqueous ammonia iii hydrogen halides with ammonia gas and with water
8.28	be able to make predictions about fluorine and astatine and their compounds, in terms of knowledge of trends in halogen chemistry

The topic is built from three sources: **8A Redox Chemistry** (8.1 – 8.10), **8B The Elements of Groups 1 and 2** (8.11 – 8.23) and **8C Inorganic Chemistry of Group 7** (8.24 – 8.28). Each assessment statement has its own section below, in specification order.

SPEC 8.1 ≈ 14 MIN

Oxidation Numbers and the Rules

THE SPECIFICATION SAYS

know what is meant by the term 'oxidation number' and understand the rules for assigning oxidation numbers

What an oxidation number is — and the rules for assigning one.

DEFINITION

Oxidation number

The charge on an atom of the element in a compound calculated assuming that all the atoms in the compound are simple monoatomic ions.

Oxidation numbers are assigned to elements in covalent compounds so that the oxidation number of an atom is the charge it would have if the electrons in each of its bonds belonged to the more electronegative atom.

Oxidation numbers are used to tell whether oxidation or reduction has taken place.

- In **MgO** the oxidation number of magnesium is **+2** since the charge on the magnesium ion is 2+. Similarly the oxidation number of oxygen is **−2**.
- In **SO_2** the oxidation number of sulfur is **+4** because if the compound were fully ionic the sulfur ion would have a charge of 4+. The oxidation number of oxygen is once again **−2**.

Here are the rules for determining oxidation number:

ATOM	OXIDATION NUMBER	EXAMPLE
Uncombined element	0	O in O_2 = 0
Simple monoatomic ion	charge on ion	Na in Na^+ = +1
Group 1, 2 & 3 element in a compound	+1, +2 & +3 respectively	Na = +1; Mg = +2
Fluorine in a compound	−1	F in BF_3 = −1
Hydrogen in a compound (except in metal hydrides, where it is −1)	+1	H in HCl = +1; H in NaH = −1
Oxygen in a compound (except in F_2O, peroxides and superoxides)	−2	O in MgO = −2; O in F_2O = +2; peroxide: −1 (e.g. K_2O_2); superoxide: −½ (e.g. KO_2)
Halogens	−1	Cl in HCl = −1
Other halogens with fluorine	+1	Cl in ClF = +1
Halogens (except fluorine) with oxygen	+1, +3, +5 & +7	Cl in ClO^- = +1; Cl in ClO_3^- = +5

ATOM	OXIDATION NUMBER	EXAMPLE
Sum of the oxidation numbers of the elements in a compound or molecule	0	In CO_2 the oxidation number of C is +4 and of O is −2, i.e. +4 + (2 × −2) = 0
Sum of the oxidation numbers of the elements in a polyatomic ion	charge on the ion	In SO_4^{2-} the oxidation number of S is +6 and of O is −2, i.e. +6 + (4 × −2) = −2

SPEC 8.2 ≈ 14 MIN

Calculating Oxidation Numbers

THE SPECIFICATION SAYS

be able to calculate the oxidation number of elements in compounds and ions, including in peroxides and metal hydrides

Calculating oxidation numbers in compounds and ions, including peroxides and metal hydrides.

To calculate an unknown oxidation number, fill in the elements whose oxidation numbers you know from the rules, then use the fact that the oxidation numbers add up to **0** in a compound, or to the **charge on the ion**.

WORKED EXAMPLE **Oxidation numbers of chlorine**

PROBLEM

Deduce the oxidation number of chlorine in: (a) NaCl (b) NaClO (c) $NaClO_3$.

1. (a) Na is +1: +1 + x = 0, so **x = −1**.
2. (b) Na is +1 and O is −2: +1 + x + (−2) = 0, so **x = +1**.
3. (c) Na is +1 and 3 O at −2 give −6: +1 + x + (−6) = 0, so **x = +5**.

ANSWER **Cl is −1 in NaCl, +1 in NaClO and +5 in $NaClO_3$.**

WORKED EXAMPLE **Oxidation numbers of nitrogen**

PROBLEM

Deduce the oxidation number of nitrogen in: (a) NH_3 (b) NO_2^- (c) NO_3^-.

1. (a) H is +1: x + 3(+1) = 0, so **x = −3**.
2. (b) O is −2 and the charge on the ion is −1: x + 2(−2) = −1, so **x = +3**.
3. (c) x + 3(−2) = −1, so **x = +5**.

ANSWER **N is −3 in NH_3, +3 in NO_2^- and +5 in NO_3^-.**

Peroxides, superoxides, metal hydrides and F_2O

WORKED EXAMPLE **Calculating the exceptions**

PROBLEM

Find the oxidation number of: (a) O in H_2O_2 (b) O in K_2O_2 (c) O in KO_2 (d) H in NaH (e) O in F_2O.

1. (a) H is +1: 2(+1) + 2x = 0, so **x = −1**.
2. (b) K is +1: 2(+1) + 2x = 0, so **x = −1** — in a **peroxide** oxygen is −1.
3. (c) K is +1: +1 + x = 0, so **x = −½** — in a **superoxide** oxygen is −½.
4. (d) Na is +1: +1 + x = 0, so **x = −1** — in a **metal hydride** hydrogen is −1.
5. (e) F is −1: x + 2(−1) = 0, so **x = +2**.

ANSWER **O is −1 in the peroxides, −½ in the superoxide and +2 in F_2O; H is −1 in the metal hydride.**

SPEC 8.3 ≈ 10 MIN

Systematic Names and Roman Numerals

THE SPECIFICATION SAYS

be able to indicate the oxidation number of an element in a compound or an ion, using a Roman numeral

Iron(II) or iron(III)? Manganate(VII) — the numbers in the brackets.

Oxidation numbers are often included in naming compounds and ions that contain elements with varying oxidation numbers. This name is often referred to as a **systematic name**.

- If the **cation** shows variable oxidation state, the Roman numeral is written in brackets following the name of the cation. **Manganese(IV) oxide** shows Mn is in the +4 oxidation state.
- If the **anion** contains an element of variable oxidation state, the Roman numeral is written in brackets following the name of the anion. **Potassium manganate(VII)** shows Mn is in the +7 oxidation state.
- Names of **oxoanions** end with the suffix **-ate**.

FORMULA OF COMPOUND OR ION	RELEVANT OXIDATION NUMBER	SYSTEMATIC NAME OF COMPOUND OR ION
$FeCl_2$	Fe +2	iron(II) chloride
$FeCl_3$	Fe +3	iron(III) chloride
$KMnO_4$	Mn +7	potassium manganate(VII)
K_2MnO_4	Mn +6	potassium manganate(VI)
CrO_4^{2-}	Cr +6	chromate(VI)

FORMULA OF COMPOUND OR ION	RELEVANT OXIDATION NUMBER	SYSTEMATIC NAME OF COMPOUND OR ION
$Cr_2O_7^{2-}$	Cr +6	chromate(VI)

Sodium and magnesium have only one oxidation number in their compounds (+1 and +2 respectively), so it is not necessary to refer to sodium chloride as sodium(I) chloride or magnesium oxide as magnesium(II) oxide.

SPEC 8.4 ≈ 8 MIN

Writing Formulae from Oxidation Numbers

THE SPECIFICATION SAYS

be able to write formulae given oxidation numbers

Read the name, balance the charges, write the formula.

It is equally important to be able to work out the formula of the compound or ion by looking at the systematic name with its oxidation number: take the oxidation numbers from the name, then choose the numbers of each ion that make the total charge zero.

WORKED EXAMPLE **Formulae of the iron sulfates**

PROBLEM

Deduce the formula of (a) iron(II) sulfate (b) iron(III) sulfate.

1. (a) Iron(II) is Fe^{2+} and sulfate is SO_4^{2-}. One of each balances the charges (+2 and −2): **$FeSO_4$**.
2. (b) Iron(III) is Fe^{3+}. Two Fe^{3+} (+6) balance three SO_4^{2-} (−6): **$Fe_2(SO_4)_3$**.

ANSWER **$FeSO_4$ and $Fe_2(SO_4)_3$.**

SYSTEMATIC NAME	IONS INVOLVED	FORMULA
copper(I) oxide	Cu^+ and O^{2-}	Cu_2O
copper(II) oxide	Cu^{2+} and O^{2-}	CuO
chromium(III) sulfate(VI)	Cr^{3+} and SO_4^{2-}	$Cr_2(SO_4)_3$
lead(IV) iodide	Pb^{4+} and I^-	PbI_4
cobalt(III) nitrate(V)	Co^{3+} and NO_3^-	$Co(NO_3)_3$

SPEC 8.5 ≈ 14 MIN

Oxidation, Reduction and Electron Transfer

> THE SPECIFICATION SAYS
>
> **understand oxidation and reduction in terms of electron transfer and changes in oxidation number, and the application of these ideas to reactions of s-block and p-block elements**

From oxygen transfer to electron transfer — and to changes in oxidation number.

Background to oxidation and reduction

'Oxidation' is known as the addition of oxygen and 'reduction' is known as the removal of oxygen.

An example of such a reaction is the reaction between iron(III) oxide and carbon monoxide that takes place in the blast furnace.

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

Iron(III) oxide loses oxygen (reduction); carbon monoxide gains oxygen (oxidation).

Oxidation can also be described as removal of hydrogen and reduction can be described as addition of hydrogen.

An example of such a reaction is the reaction between chlorine and hydrogen sulfide.

$$H_2S + Cl_2 \rightarrow S + 2HCl$$

H_2S loses hydrogen (oxidation); Cl_2 gains hydrogen (reduction).

Oxidation and reduction in terms of electron loss and gain

Oxidation and reduction involve the transfer of electrons.

When magnesium burns in oxygen it forms magnesium oxide.

$$2Mg + O_2 \rightarrow 2MgO$$

The reaction results in the formation of Mg^{2+} and O^{2-} ions.

$$Mg \rightarrow Mg^{2+} + 2e^-$$

Each of the two magnesium atoms has lost two electrons.

$$O_2 + 4e^- \rightarrow 2O^{2-}$$

The oxygen molecule has gained four electrons to become oxide ions.

The magnesium has been **oxidised** because it has **lost** electrons. Oxygen has been **reduced** because it has **gained** electrons.

We now have new definitions of oxidation and reduction:

DEFINITION

Oxidation

Oxidation is the **loss** of electrons.

DEFINITION

Reduction

Reduction is the **gain** of electrons.

KEY POINT

OIL RIG

These definitions are easily remembered using the mnemonic **OIL RIG**: **O**xidation **I**s **L**oss (of electrons), **R**eduction **I**s **G**ain (of electrons).

Redox reaction

- A redox reaction is a chemical reaction in which oxidation and reduction occur at the same time.
- Electrons are transferred in such reactions.
- The reaction between zinc and copper(II) sulfate solution is an example of a redox reaction.

In the following reaction, zinc metal gives up electrons:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

Oxidation

The electrons are then taken up by the copper ions:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

Reduction

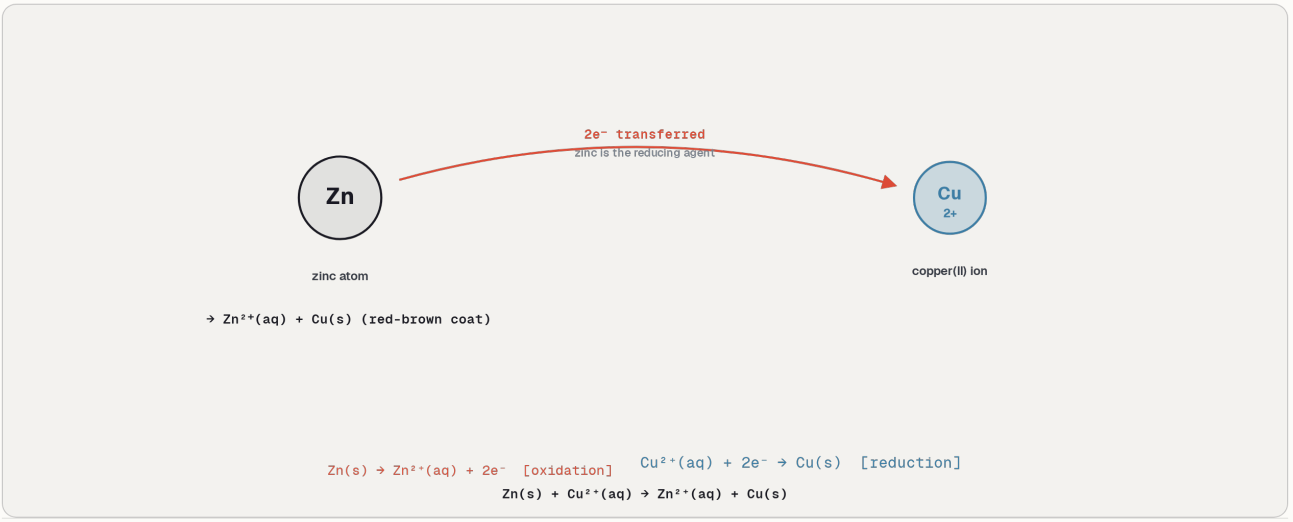
$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

The two half-equations added together — the electrons cancel.

The overall equation for the reaction is:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

SPEC 8.5 · 8.6 · ELECTRON TRANSFER **Zinc displacing copper from copper(II) sulfate** redox = transfer



MODEL Each zinc atom gives two electrons directly to a copper(II) ion. The half-equations are added with the electrons cancelling — electrons never appear in the overall ionic equation.

FIG — *Electron transfer in the reaction between zinc and copper(II) sulfate solution.*

SPEC 8.6 ≈ 10 MIN

Oxidising and Reducing Agents

THE SPECIFICATION SAYS

know that oxidising agents gain electrons and reducing agents lose electrons

Oxidising agents gain electrons; reducing agents lose electrons.

- An **oxidising agent** is a species (atom, molecule or ion) that oxidises another species by removing one or more electrons and itself gets reduced.
- Non-metals are oxidising agents, e.g. F_2, Cl_2, O_2.
- A **reducing agent** is a species that reduces another species by giving it one or more electrons and itself gets oxidised.
- Metals are reducing agents, e.g. Na, Fe, Zn.

DEFINITION

Oxidising agent

It is an **electron acceptor**. It oxidises another substance. The oxidation number of the element in it **decreases**.

DEFINITION

Reducing agent

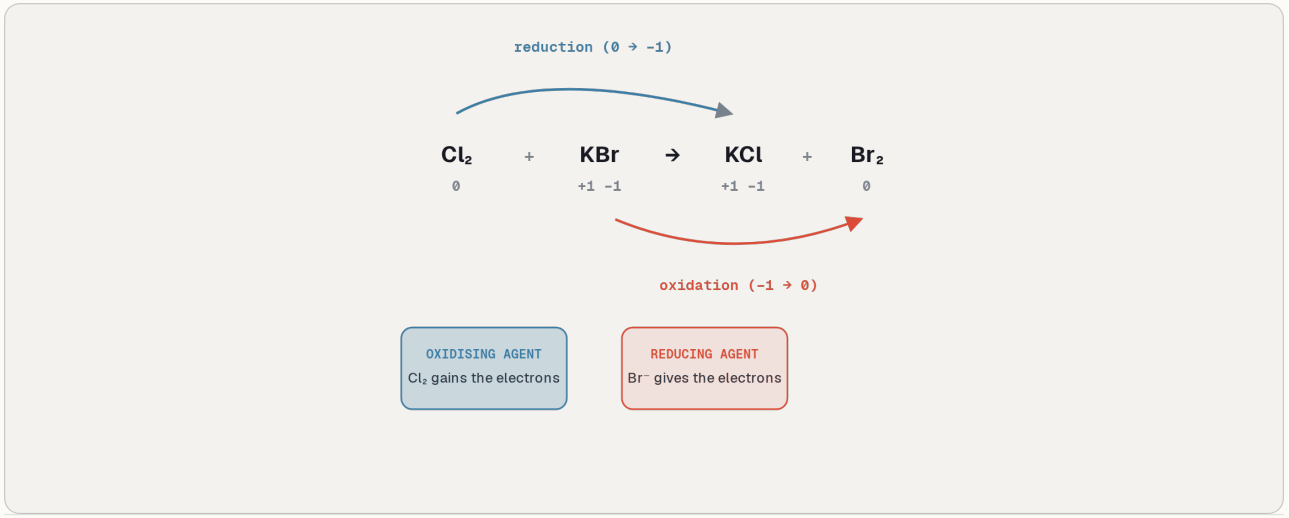
It is an **electron donor**. It reduces another substance. The oxidation number of the element in it **increases**.

In the reaction between chlorine and potassium bromide:

$$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$$

Cl_2 is reduced (0 → −1) — the oxidising agent. Br^- is oxidised (−1 → 0) — the reducing agent.

SPEC 8.6 · NAMING THE AGENTS **Oxidising and reducing agents in Cl_2 + KBr** · oxidising agent: electron acceptor · reducing agent: electron donor



READ IT OFF The oxidising agent (Cl_2) is itself reduced; the reducing agent (Br^-) is itself oxidised.

FIG — *Cl_2 gains the electrons (oxidising agent, itself reduced); Br^- gives the electrons (reducing agent, itself oxidised).*

SPEC 8.7 ≈ 12 MIN

Disproportionation

THE SPECIFICATION SAYS

understand that a disproportionation reaction involves an element in a single species being simultaneously oxidised and reduced

One element, one species — oxidised and reduced at the same time.

DEFINITION

Disproportionation

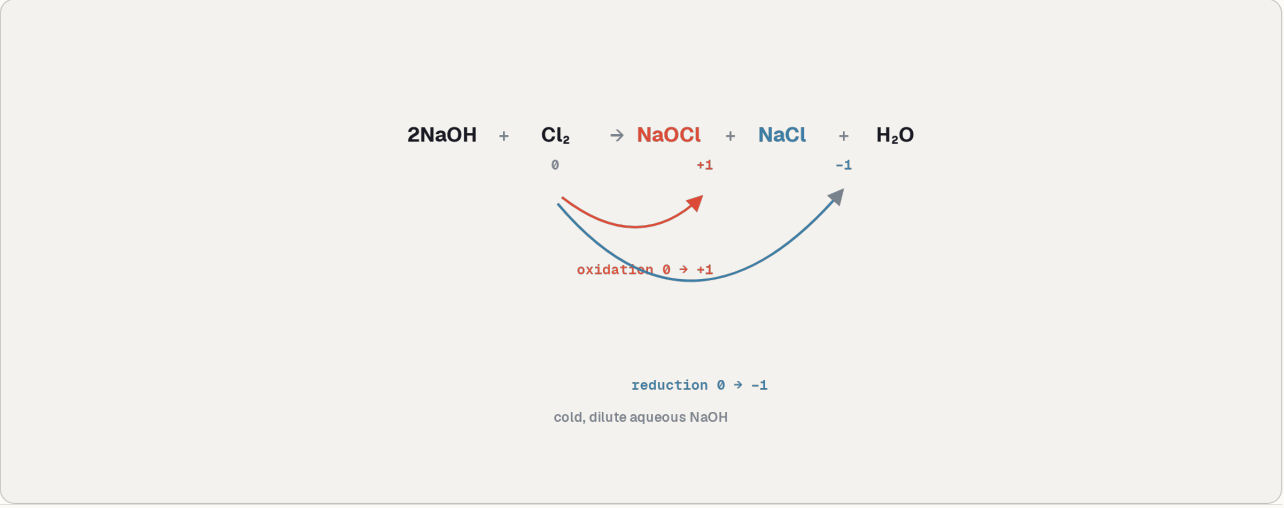
A disproportionation reaction takes place when the same element is both oxidised and reduced **simultaneously**.

Example: when chlorine gas is passed into aqueous sodium hydroxide, a disproportionation reaction takes place. One atom of chlorine is oxidised and the other is reduced.

$$2NaOH + Cl_2 \rightarrow NaOCl + NaCl + H_2O$$

Cl: 0 → +1 in NaOCl (oxidised); Cl: 0 → −1 in NaCl (reduced).

SPEC 8.7 · DISPROPORTIONATION **Chlorine in cold dilute sodium hydroxide** — one element · one species · both ways



THE TEST Both chlorine atoms start at 0 inside Cl_2. One climbs to +1 (sodium chlorate(I)), the other falls to −1 (sodium chloride). Oxidised and reduced simultaneously.

FIG — *One atom of chlorine is oxidised and the other is reduced.*

WORKED EXAMPLE — Heating potassium bromate(I)

PROBLEM

When potassium bromate(I) is heated it decomposes: $3KOBr \rightarrow 2KBr + KBrO_3$. Using the oxidation numbers of the element involved, explain why this is a disproportionation reaction.

1. Br in KOBr is **+1**; Br in KBr is **−1**; Br in $KBrO_3$ is **+5**.
2. Bromine starts at +1 in the single species KOBr and finishes at −1 and +5.
3. Bromine has been reduced (+1 → −1) and oxidised (+1 → +5) at the same time.

ANSWER **The same element (Br), in a single species (KOBr), is simultaneously oxidised and reduced — disproportionation.**

WORKED EXAMPLE — Decomposition of hydrogen peroxide

PROBLEM

Is the decomposition $2H_2O_2 \rightarrow 2H_2O + O_2$ a disproportionation reaction? Justify your answer.

1. O in H_2O_2 is **−1** (a peroxide); O in H_2O is **−2**; O in O_2 is **0**.
2. Oxygen starts at −1 in the single species H_2O_2 and finishes at −2 and 0.
3. Oxygen has been reduced (−1 → −2) and oxidised (−1 → 0) at the same time.

ANSWER **Yes — oxygen in H_2O_2 is simultaneously oxidised and reduced, so the decomposition is a disproportionation reaction.**

SPEC 8.8 ≈ 8 MIN

Classifying Reactions with Oxidation Numbers

> THE SPECIFICATION SAYS
>
> know that oxidation number is a useful concept in terms of the classification of reactions as redox and as disproportionation

Assign the numbers, look for changes, name the reaction.

Using oxidation numbers to classify reactions

Consider the following ionic half-equations:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

$$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$$

$$2I^-(aq) \rightarrow I_2(aq) + 2e^-$$

Electrons have been lost, so the reactions are oxidations. The oxidation number of the element has increased in each case: Zn from **0 to +2**, Fe from **+2 to +3**, I from **−1 to 0**.

Another definition of oxidation and reduction:

- **Oxidation** occurs when there is an **increase** in oxidation number.
- **Reduction** occurs when there is a **decrease** in oxidation number.

Now consider the following ionic half-equations:

$$Cl_2(aq) + 2e^- \rightarrow 2Cl^-(aq)$$

Cl changes from 0 to −1.

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$$

Mn changes from +7 to +2.

$$2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g)$$

H changes from +1 to 0.

In each case the oxidation number of one of the elements involved has decreased, so reduction has taken place.

WORKED EXAMPLE **Redox or not?**

PROBLEM

Use oxidation numbers to show whether the following reactions are examples of redox reactions: (a) $H_2S(g) + Cl_2(g) \rightarrow 2HCl(g) + S(s)$ (b) $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$ (c) $2NaOH(aq) + Cl_2(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$.

1. (a) Cl: 0 → −1 (decrease, reduced); S: −2 → 0 (increase, oxidised). Oxidation and reduction occur at the same time → **redox**.
2. (b) Na is +1, O is −2, H is +1, Cl is −1 — no element changes its oxidation number → **not redox**.
3. (c) Cl: 0 → −1 in NaCl and 0 → +1 in NaOCl. The same element is oxidised and reduced → **redox, and a disproportionation**.

ANSWER **(a) redox; (b) not redox; (c) redox — and a disproportionation.**

SPEC 8.9 ≈ 10 MIN

Metals, Non-metals and Oxidation Number

THE SPECIFICATION SAYS

understand that metals, in general, form positive ions by loss of electrons with an increase in oxidation number whereas non-metals, in general, form negative ions by gain of electrons with a decrease in oxidation number

Metals form positive ions by electron loss; non-metals form negative ions by electron gain.

Reactions of s-block elements

Reactive metals in groups 1 and 2 lose electrons when they react.

ELEMENT	ELECTRONS LOST	HALF EQUATION	CHANGE IN OXIDATION NUMBER
Sodium (Na) — Group 1	1	$Na \rightarrow Na^+ + e^-$	increase in oxidation number from 0 to +1
Magnesium (Mg) — Group 2	2	$Mg \rightarrow Mg^{2+} + 2e^-$	increase in oxidation number from 0 to +2

All group 1 and 2 elements undergo oxidation.

Reactions of p-block elements

Predictions about the reactions of p-block elements can be made in terms of redox reactions. However it is not as straightforward as for s-block elements, because the p-block contains elements with different characteristics.

- The group 3 element **aluminium** is a metal with reactivity nearly as high as magnesium, so it reacts in a similar way: $Al \rightarrow Al^{3+} + 3e^-$ — an **increase** in oxidation number from 0 to +3.

- The group 7 element **fluorine** is the most reactive halogen: $F_2 + 2e^- \rightarrow 2F^-$ — a **decrease** in oxidation number from 0 to −1.
- **Nitrogen** is a p-block element in group 5, with an electronegativity higher than aluminium but lower than fluorine. The change in oxidation number for nitrogen therefore depends on which element it reacts with.

Nitrogen reacts with the group 1 metal sodium: $6Na + N_2 \rightarrow 2Na_3N$. In this redox reaction the oxidation number of sodium increases from 0 to +1 and that of nitrogen decreases from 0 to −3.

Now consider the reaction of nitrogen with fluorine: $N_2 + 3F_2 \rightarrow 2NF_3$. The oxidation number of fluorine decreases from 0 to −1, but this time the oxidation number of nitrogen increases from 0 to +3.

$$6Na + N_2 \rightarrow 2Na_3N \qquad N_2 + 3F_2 \rightarrow 2NF_3$$

With sodium, nitrogen is reduced (0 → −3); with fluorine, nitrogen is oxidised (0 → +3).

SPEC 8.10 ≈ 30 MIN

Ionic Half-Equations and Full Ionic Equations

THE SPECIFICATION SAYS

be able to write ionic half-equations and use them to construct full ionic equations

Balance atoms, balance charge with electrons, then combine — in acid or alkali.
A half equation represents either an oxidation or a reduction.

- **Oxidation** half equations have electrons on the **right hand side**: $X \rightarrow X^+ + 1e^-$.
- **Reduction** half equations have electrons on the **left hand side**: $X + 1e^- \rightarrow X^-$.

Balancing simple half equations

1. Write down the formula of the reactants and products.
2. Balance the atoms.
3. Add electrons to the right hand side if it is oxidation, and to the left hand side if it is reduction.
4. Change the number of electrons to balance the charge on both sides of the equation.

STEP	IRON(II) BEING OXIDISED TO IRON(III)	CHLORINE BEING REDUCED TO CHLORIDE
1	$Fe^{2+} \rightarrow Fe^{3+}$	$Cl_2 \rightarrow Cl^-$
2	$Fe^{2+} \rightarrow Fe^{3+}$ (atom already balanced)	$Cl_2 \rightarrow 2Cl^-$
3	$Fe^{2+} \rightarrow Fe^{3+} + e^-$	$Cl_2 + e^- \rightarrow 2Cl^-$
4	$Fe^{2+} \rightarrow Fe^{3+} + e^-$ (already balanced)	$Cl_2 + 2e^- \rightarrow 2Cl^-$

Balancing half equations in acidic medium

1. Write down the formula of the reactants and products.
2. Balance the atoms other than oxygen and hydrogen.
3. Balance oxygen atoms by adding H_2O to the side which requires oxygen atoms.
4. Balance hydrogen atoms by adding H^+ to the side which requires hydrogen atoms.
5. Add electrons to balance the charge on both sides of the equation.

Example: MnO_4^- being reduced to Mn^{2+} in acidic medium.

STEP	HALF-EQUATION
1	$MnO_4^- \rightarrow Mn^{2+}$
2	$MnO_4^- \rightarrow Mn^{2+}$ (Mn balanced)
3	$MnO_4^- \rightarrow Mn^{2+} + 4H_2O$
4	$MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$
5	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ — +7 on the left side and +2 on the right, so the charge is balanced by adding $5e^-$ to the left side

Balancing half equations in alkaline medium

1. Write down the formula of the reactants and products.
2. Balance the atoms other than oxygen and hydrogen.
3. Add OH^- to the side which requires oxygen atoms, and double the number of OH^- to the required number of oxygen atoms.
4. Balance hydrogen atoms by adding H_2O to the side which requires hydrogen atoms.
5. Add electrons to balance the charge on both sides of the equation.

Example: MnO_4^- being reduced to Mn^{2+} in alkaline medium.

STEP	HALF-EQUATION
1	$MnO_4^- \rightarrow Mn^{2+}$
2	$MnO_4^- \rightarrow Mn^{2+}$ (Mn balanced)
3	$MnO_4^- \rightarrow Mn^{2+} + 8OH^-$ — 4 oxygen atoms on the left side, so $4 \times 2 = 8$ OH^-
4	$MnO_4^- + 4H_2O \rightarrow Mn^{2+} + 8OH^-$
5	$MnO_4^- + 4H_2O + 5e^- \rightarrow Mn^{2+} + 8OH^-$ — −1 on the left side and −6 on the right, so the charge is balanced by adding $5e^-$ to the left side

Writing balanced overall equations

1. Write the balanced half equation for oxidation and for reduction.
2. Multiply the equations so that the number of electrons in each is the same.
3. Add the equations and cancel out the electrons on either side of the equation.
4. If necessary, cancel out any other species which appear on both sides.

Example: the reaction between manganate(VII) and iron(II).

STEP	EQUATIONS
1	$Fe^{2+} \rightarrow Fe^{3+} + e^-$ (oxidation) · $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (reduction)
2	$5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$ (multiply by 5) · $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (multiply by 1)
3	$5Fe^{2+} + MnO_4^- + 8H^+ + 5e^- \rightarrow 5Fe^{3+} + 5e^- + Mn^{2+} + 4H_2O$ (add the equations, then cancel the electrons)

$$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$$

The overall equation — the electrons have cancelled.

SPEC 8.11 ≈ 12 MIN

Ionisation Energy Trends in Groups 1 and 2

THE SPECIFICATION SAYS

understand reasons for the trend in ionisation energy down Groups 1 and 2

The Group 1 and 2 elements, their configurations, atomic radii and the fall in ionisation energy down the group.

Introduction to the Group 1 and 2 elements

- The **alkali metals** and **alkaline-earth metals** are members of Group I and Group II, respectively, in the periodic table.
- They are **solids at room temperature**.
- The pure elements in these groups (except Francium and Radium) appear as **bright silvery solids**. However, when exposed to air, they combine with oxygen to form oxides as surface layers, which makes them appear dull.
- They are **s-block elements** since their valence electrons are in s orbitals.
- All Group I metals have **one outermost s-electron**, whereas Group II metals have **two** outermost s-electrons. These electrons are situated much further from the nucleus than all other electrons in the metals, so they are only weakly held by the positive nucleus. As a result, Group I and II elements readily lose their outermost s-electrons to form stable ions with **noble gas structures**.

GROUP I – THE ALKALI METALS	GROUP II – THE ALKALINE-EARTH METALS
Lithium Li	Beryllium Be
Sodium Na	Magnesium Mg
Potassium K	Calcium Ca
Rubidium Rb	Strontium Sr
Caesium Cs	Barium Ba

GROUP I	ELECTRONIC CONFIGURATION	GROUP II	ELECTRONIC CONFIGURATION
Li Lithium	[He]$2s^1$	Be Beryllium	[He]$2s^2$
Na Sodium	[Ne]$3s^1$	Mg Magnesium	[Ne]$3s^2$
K Potassium	[Ar]$4s^1$	Ca Calcium	[Ar]$4s^2$
Rb Rubidium	[Kr]$5s^1$	Sr Strontium	[Kr]$5s^2$
Cs Caesium	[Xe]$6s^1$	Ba Barium	[Xe]$6s^2$
***Fr Francium**	[Rn]$7s^1$	*Ra Radium	[Rn]$7s^2$

Atomic radius

- Atomic radius **increases** down the Group I and II as the outer most electrons are further away from the nuclei.
- Atomic radius of Group II < Group I.

GROUP 1 ELEMENT	ATOMIC RADIUS / NM	GROUP II ELEMENT	ATOMIC RADIUS / NM
Li	0.152	Be	0.112
Na	0.186	Mg	0.160
K	0.231	Ca	0.197
Rb	0.244	Sr	0.215
Cs	0.262	Ba	0.224

Ionisation energy

- **Ionisation energy decreases** down the Group I and II.
- Down the group, the atomic size increases and the number of shells increases.
- Distance between the nucleus and the outer electrons increases.
- The outer electrons are more shielded by the electrons in the inner shells.
- The effect of the increase in nuclear charge is outweighed by the increase in shielding.

- Therefore, the outer electrons are less attracted by the nucleus.
- Hence, less energy is required to remove the electrons.

GROUP I ELEMENT	1ST IE / KJ MOL⁻¹	2ND IE / KJ MOL⁻¹	GROUP II ELEMENT	1ST IE / KJ MOL⁻¹	2ND IE / KJ MOL⁻¹	1ST AND 2ND IE / KJ MOL⁻¹
Li	519	7 300	Be	900	1 757	2 657
Na	494	4 560	Mg	738	1 451	2 189
K	418	3 070	Ca	590	1 145	1 735
Rb	402	2 370	Sr	550	1 064	1 614
Cs	376	2 420	Ba	503	965	1 468

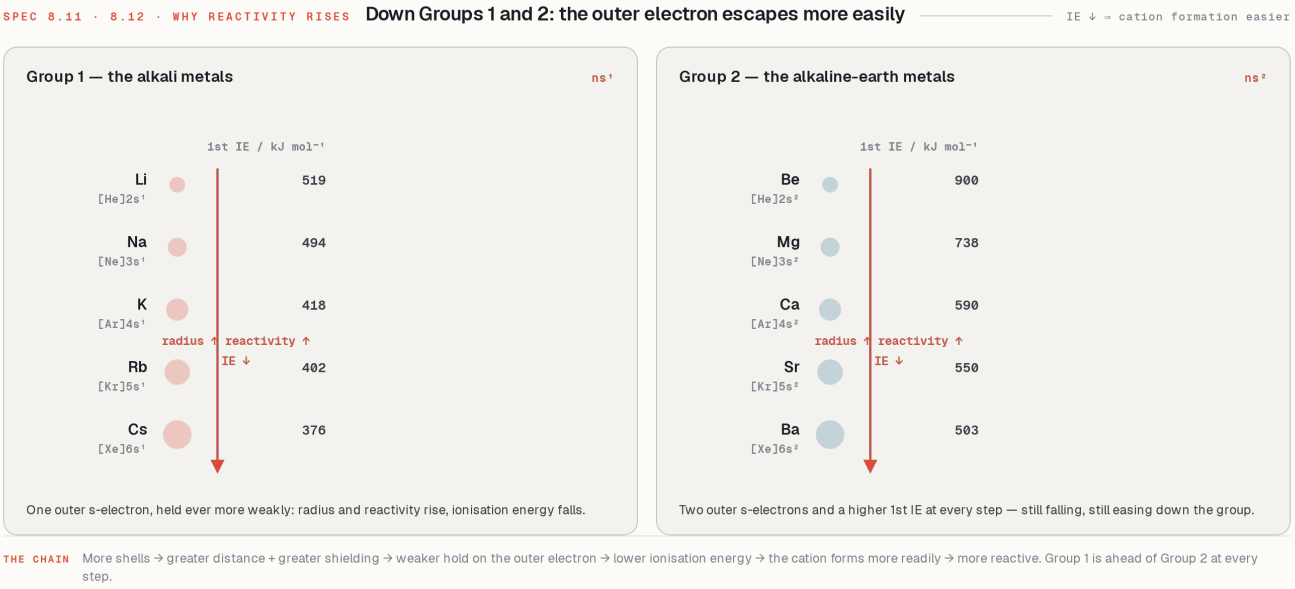


FIG — *Down the group the outer electron is held more weakly — radius and reactivity rise, ionisation energy falls.*

SPEC 8.12 ≈ 10 MIN

Reactivity Trends in Groups 1 and 2

> THE SPECIFICATION SAYS
>
> understand reasons for the trend in reactivity of the elements down Group 1 (Li to K) and Group 2 (Mg to Ba)

Reactivity increases down the group — and Group I is more reactive than Group II.

Reactivity

- The reactivity of elements **increases down the group** because of the tendency of elements to lose electron increases down the group. This is because ionization energy decreases, meaning **less energy is required to form the cations**.

- **Group I elements are more reactive than Group II** because Group I elements can lose electrons easily and are more metallic.

SPEC 8.13 ≈ 24 MIN

Reactions with Oxygen, Chlorine and Water

> THE SPECIFICATION SAYS
>
> know the reactions of the elements of Group 1 (Li to K) and Group 2 (Mg to Ba) with oxygen, chlorine and water

Reactions of the Group 1 and 2 elements with oxygen, chlorine and water, with the observations for each metal.

Oxygen

- S-block elements react readily with oxygen to form **solid metal oxides**.

	GROUP I METAL WITH OXYGEN	GROUP II METAL WITH OXYGEN
General equation	$4M(s) + O_2(g) \rightarrow 2M_2O(s)$	$2M(s) + O_2(g) \rightarrow 2MO(s)$
Products	products are oxides containing M^+ and O^{2-} ions	products are oxides containing M^{2+} and O^{2-} ions

Chlorine

- All s-block elements react with chlorine to form **ionic chlorides** except beryllium.
- The ionic chlorides of Group I and II are **soluble in water**.

	GROUP I METAL WITH CHLORINE	GROUP II METAL WITH CHLORINE
General equation	$2M(s) + Cl_2(g) \rightarrow 2MCl(s)$	$M(s) + Cl_2(g) \rightarrow MCl_2(s)$
Products	products are chlorides containing M^+ and Cl^- ions	products are chlorides containing M^{2+} and Cl^- ions

Water

- S-block elements react with water to produce **alkaline metal hydroxide** and **hydrogen**.
- **Group 1:** the general equation is

$$2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$$

Group 1 metals with water

Lithium

- When a small piece of lithium is added to a beaker or trough of water, it **floats** on the surface of the water (density is half of that of water), gently fizzing and giving off hydrogen.
- It gradually reacts and disappears forming a colorless solution of lithium hydroxide.

$$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$$

Lithium with cold water

Sodium

- When sodium is added to water, the sodium **melts to form a ball** that moves around on the surface. It fizzes rapidly, and the hydrogen produced may burn with an **orange flame** (the colour is due to the contamination of hydrogen flame with sodium compounds) before the sodium disappears.
- The flame is caused by the heat of the reaction igniting the hydrogen.

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

Sodium with cold water

Potassium

- Potassium behaves rather like sodium except that the reaction is **faster** and enough heat is given off to set light to the hydrogen.
- When potassium is added to water, the metal melts and floats. It moves around very quickly on the surface of the water. The hydrogen ignites instantly. The metal is also set on fire, with sparks and a **lilac flame**.

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

Potassium with cold water

- **Group II:** the general equation for all these reactions is

$$M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq) + H_2(g)$$

Group 2 metals with cold water

Magnesium

- Reacts **very slowly** with cold water to form magnesium hydroxide and hydrogen gas.

$$Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(s) + H_2(g)$$

Magnesium with cold water

Magnesium and steam

- Magnesium reacts differently when heated in steam.
- It rapidly forms magnesium oxide (a white solid) and hydrogen gas in a **vigorous** reaction. The equation for the reaction is

$$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$$

Magnesium with steam

- The hydrogen formed is burned as it leaves the tube. This is for safety reasons to prevent the escape of highly flammable gas into the laboratory.

Calcium, Strontium and Barium

- React **rapidly** with cold water to form metal hydroxide and hydrogen gas.

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(s) + H_2(g)$$

Calcium with cold water

$$Sr(s) + 2H_2O(l) \rightarrow Sr(OH)_2(aq) + H_2(g) \qquad Ba(s) + 2H_2O(l) \rightarrow Ba(OH)_2(aq) + H_2(g)$$

Strontium and barium with cold water

SPEC 8.14 ≈ 14 MIN

Oxides and Hydroxides with Water and Acids

THE SPECIFICATION SAYS

know the reactions of: i oxides of Group 1 and 2 elements with water and dilute acid ii hydroxides of Group 1 and 2 elements with dilute acid

Basic oxides (beryllium oxide is amphoteric) and their reactions with water and with dilute acids.

Oxides

- Group one and two oxides are classified as **basic oxides** (beryllium oxide/hydroxide is **amphoteric**).
- They react with water to form **alkaline hydroxides**.

Reactions of Group I oxides with water

GROUP I OXIDE	NATURE OF OXIDE	REACTION WITH WATER
M_2O (in general)	Basic	$M_2O(s) + H_2O(l) \rightarrow 2MOH(aq)$

Reactions of Group II oxides with water

GROUP II OXIDE	NATURE OF OXIDE	REACTION WITH WATER
MO (in general)	Basic	$MO(s) + H_2O(l) \rightarrow M(OH)_2(aq)$
MgO(s)	Basic	Reacts very slowly and incompletely to form a slightly alkaline suspension of $Mg(OH)_2$. $MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(s)$ (white ppt)
CaO(s)	Basic	$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(s)$ (cloudy suspension)

Hydroxides and oxides with acids

- All of the **Group 1 and 2 oxides and hydroxides** behave as **bases** and **react with acids** to form **salts and water**.
- These reactions can be described as **neutralization** reactions.
- During the reactions, the only observations to be made are that a **white solid** reacts to form a **colourless solution**.
- The reactions are highly **exothermic**.

SPEC 8.15 ≈ 12 MIN

Solubility Trends of Group 2 Hydroxides and Sulfates

> THE SPECIFICATION SAYS
>
> know the trends in solubility of the hydroxides and sulfates of Group 2 elements

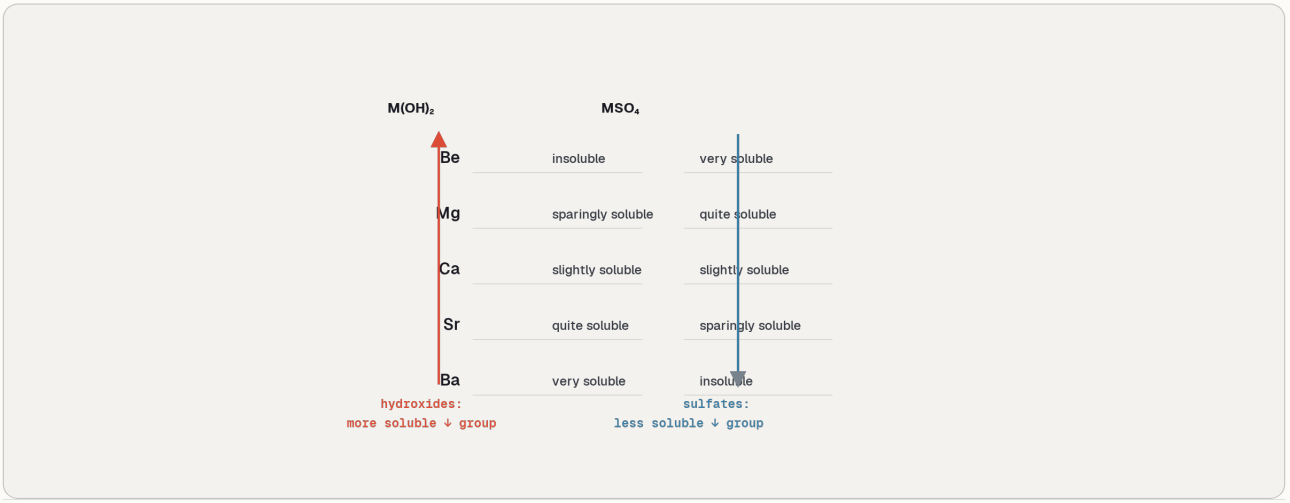
Solubility of Group 2 hydroxides increases down the group; sulfates decreases.

Solubility of Group II hydroxides and sulphates

- Solubility of **group 2 hydroxides increases** down the group.
- Solubility of **group 2 sulphates decreases** down the group.

SOLUBILITY OF HYDROXIDES		SOLUBILITY OF SULPHATES	
$Be(OH)_2$	insoluble	$BeSO_4$	very soluble
$Mg(OH)_2$	sparingly soluble	$MgSO_4$	quite soluble
$Ca(OH)_2$	slightly soluble	$CaSO_4$	slightly soluble
$Sr(OH)_2$	quite soluble	$SrSO_4$	sparingly soluble
$Ba(OH)_2$	very soluble	$BaSO_4$	insoluble

SPEC 8.15 · TWO OPPOSING LADDERS **Solubility of Group 2 hydroxides vs sulfates** know both directions



THE LADDERS Hydroxides dissolve more readily down the group ($Mg(OH)_2$ suspension → $Ba(OH)_2$ strong alkali); sulfates dissolve less ($MgSO_4$ soluble → $BaSO_4$ insoluble — the basis of the sulfate test).

FIG — *Hydroxide solubility increases $Be(OH)_2$ → $Ba(OH)_2$ down the group; sulfate solubility decreases $BeSO_4$ → $BaSO_4$.*

SPEC 8.16 ≈ 22 MIN

Thermal Stability of Nitrates and Carbonates

THE SPECIFICATION SAYS

understand the reasons for the trends in thermal stability of the nitrates and the carbonates of the elements in Groups 1 and 2 in terms of the size and charge of the cations involved

How cation size and charge control the thermal stability of the nitrates and carbonates.

Thermal stability

- **Thermal stability** is a term that indicates how stable a compound is when it is heated. If it does not decompose at all, it is very thermally stable. If decompose as much as possible then it is not thermally stable.
- The stabilities of the nitrate and carbonate anions are influenced by the **charge and size of the cations** present. Smaller and more highly charged cations affect these anions more.
- In general:
- Cations with small size and high charge has a higher **polarizing power**.
- So they can **polarize the anion** more effectively, which weakens the bonds in anions.
- This assists decomposition and makes the compound **less stable** to heat.

Trends in thermal stability of Group I and II nitrates and carbonates

- **Thermal stability increases down the group.** This is because:
- Size of the cation increases and charge density decreases.
- So polarizing power decreases.

- The cation cannot polarize the anion effectively.
- So the bonds in the anion are not weakened.
- As the degree of covalency decreases, the strength of the ionic lattice increases.
- Therefore, more energy is required to break the lattice structure.
- **Thermal stability decreases across the period from Group I to II.** This is because:
- Group 2 cations are smaller in size than group 1 cations.
- Charge on group 2 cations are higher.
- So, polarizing power of group 2 cations are larger.
- They can polarize the anion more effectively, weakening the bonds in it.
- So less energy is required to break the bonds.

GROUP 1 ELEMENT	CHARGE ON ION	IONIC RADIUS / NM	GROUP 2 ELEMENT	CHARGE ON ION	IONIC RADIUS / NM
Lithium	+1	0.074	Beryllium	+2	0.027
Sodium	+1	0.102	Magnesium	+2	0.072
Potassium	+1	0.138	Calcium	+2	0.100
Rubidium	+1	0.149	Strontium	+2	0.113
Caesium	+1	0.170	Barium	+2	0.136

The cation with the greatest influence (biggest charge and smaller size) on an anion is **Be^{2+}**, and the one with the least influence (smallest charge and largest size) is **Cs^{+}**.

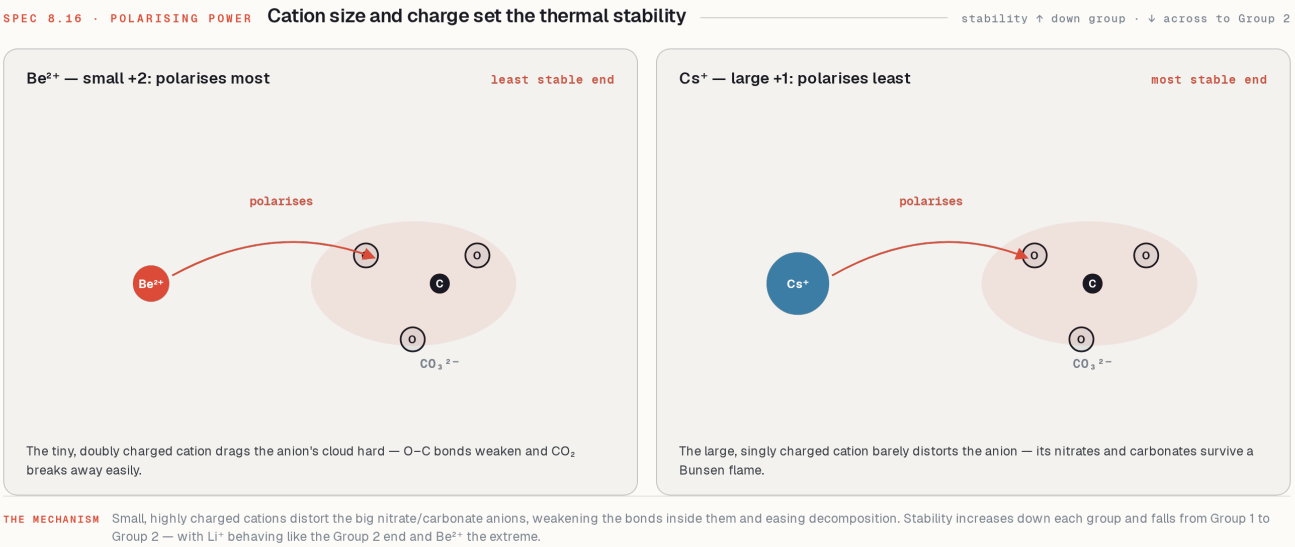


FIG — *The bond is weakened due to polarisation of the electron cloud by the cation — decomposition of the Group II nitrate occurs.*

Decomposition of Group I and II nitrates

	DECOMPOSITION OF GROUP I NITRATES	DECOMPOSITION OF GROUP II NITRATES
General equation	$2XNO_3(s) \rightarrow 2XNO_2(s) + O_2(g)$	$2X(NO_3)_2(s) \rightarrow 2XO(s) + 4NO_2(g) + O_2(g)$
Pattern	All Group I nitrates, except $LiNO_3$, decomposes to form **metal nitrites**, XNO_2 (pale yellow solid) and **oxygen gas**. $LiNO_3$ decomposes to form lithium oxide, nitrogen dioxide and oxygen.	All Group II nitrates decompose to form **metal oxide**, **nitrogen dioxide** (brown gas) and **oxygen**.
Description of nitrates	Group I nitrates are white solids.	Group II nitrates are white solids.
Equations	$4LiNO_3(s) \rightarrow 2Li_2O(s) + 4NO_2(g) + O_2(g)$ (Greater decomposition); $2NaNO_3(s) \rightarrow 2NaNO_2(s) + O_2(g)$ (Lesser decomposition)	$2Be(NO_3)_2(s) \rightarrow 2BeO(s) + 4NO_2(g) + O_2(g)$ (Greater decomposition)
Observation	$4LiNO_3(s) \rightarrow$ Brown fumes; $2NaNO_3(s)$, $2KNO_3(s)$, $2RbNO_3(s)$ and $2CsNO_3(s) \rightarrow$ No brown fumes	$2Be(NO_3)_2(s)$, $2Mg(NO_3)_2(s)$, $2Ca(NO_3)_2(s)$, $2Sr(NO_3)_2(s)$ and $2Ba(NO_3)_2(s) \rightarrow$ Brown fumes

Decomposition of Group I and II carbonates

- Group 1 carbonates are **stable to heat** except lithium carbonate, which decomposes in a similar manner to group 2 carbonates.
- Lithium carbonate decompose when heated because the **Li^+ cation is very small** and polarizes the O–C bond in the CO_3^{2-} ion sufficiently for it to break and form an O^{2-} ion.

	DECOMPOSITION OF GROUP I CARBONATES	DECOMPOSITION OF GROUP II CARBONATES
General equation	$X_2CO_3(s) \rightarrow X_2O(s) + CO_2(g)$	$XCO_3(s) \rightarrow XO(s) + CO_2(g)$
Pattern	All carbonates except lithium carbonate do not decompose at Bunsen temperature. Lithium carbonate decomposes to form lithium oxide and carbon dioxide.	All carbonates except barium carbonate decomposes to give metal oxide and carbon dioxide.
Description of carbonates	Group I carbonates are white solids.	Group II carbonates are white solids.

$$Li_2CO_3(s) \rightarrow Li_2O(s) + CO_2(g)$$

Lithium carbonate decomposes on heating — the Group I exception

- Na_2CO_3, K_2CO_3, Rb_2CO_3 and Cs_2CO_3 **do not decompose at Bunsen temperatures**.
- $BaCO_3$ **does not decompose at Bunsen temperatures**; $BeCO_3$, $MgCO_3$, $CaCO_3$ and $SrCO_3$ decompose to give the metal oxide and carbon dioxide.

SPEC 8.17 ≈ 14 MIN

Flame Colours and Electron Transitions

THE SPECIFICATION SAYS

understand the formation of characteristic flame colours by Group 1 and 2 compounds in terms of electron transitions Students will be expected to know the flame colours for Group 1 and 2 compounds.

Excitation and relaxation of electrons — and the flame colours of the Group 1 and 2 cations.

Flame test

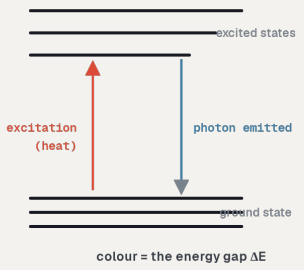
- It is an important test to identify the **metal cation** present in a compound.

Origin of flame colours

- Metal cations of the metals in Groups I and II give **characteristic colours**.
- This arises due to **electronic transitions** within the cation.
- When a compound is heated, electrons gain energy and move to higher energy levels. This is commonly referred as **excitation** of electrons.
- Excited electrons in higher energy levels are not stable. So, they fall back to the ground state, emitting energy in the form of light.
- The light emitted is of a characteristic frequency and hence colour depends on the energy level difference between the two shells.

SPEC 8.17 · ELECTRON TRANSITIONS **Excitation, relaxation, emission — the flame test** each cation has its own ΔE fingerprint

Origin of the colour 8.17



Flame colours to know CHART



Be²⁺ and Mg²⁺: no colour (ΔE emits in the UV)

PHYSICS Heat promotes cation electrons to higher levels; they fall back and emit light whose frequency matches the gap. Bigger gaps emit bluer light; Be^{2+} and Mg^{2+} gaps are so large their emission is ultraviolet.

FIG — *Heat promotes the electron to a higher energy level (excitation); it falls back to the ground state, emitting light of a characteristic frequency — the colour depends on the energy level difference between the two shells.*

Flame colours of Group I and II metal cations

METAL CATION	FORMULA	COLOUR
Lithium	Li^+	red
Sodium	Na^+	yellow/orange
Potassium	K^+	lilac
Rubidium	Rb^+	red/purple
Caesium	Cs^+	blue/violet
Beryllium	Be^{2+}	(no colour)
Magnesium	Mg^{2+}	(no colour)
Calcium	Ca^{2+}	(brick) red
Strontium	Sr^{2+}	(crimson) red
Barium	Ba^{2+}	(apple) green

SPEC 8.18 ≈ 12 MIN

Experimental Procedures: Decomposition and Flame Tests

> THE SPECIFICATION SAYS
>
> know experimental procedures to show: i patterns in the thermal decomposition of Group 1 and 2 nitrates and carbonates ii flame colours in compounds of Group 1 and 2 elements Students will be expected to know tests for carbon dioxide and oxygen; and to recognise nitrogen dioxide by its colour and acidic pH.

The thermal decomposition experiment and the flame test procedure.

Experiment to study thermal decomposition

- The relative ease of decomposition of group 2 carbonates and nitrates can be studied by the following experiment.

1. Place the same amount of each of the salts in a series of hard glass test tubes.
2. Fix a delivery tube to the test tube and clamp in a stand.
3. Light the burner and measure the time taken for the gas evolved to reach the mark on the test tube in the water bath.
4. Repeat the experiments by changing the salts.

Flame test procedure

1. Wear a safety glasses and a lab coat. Within a fume cupboard light a Bunsen burner.

2. Using a dropper, add a few drops of concentrated hydrochloric acid to the solid and mix together so that the metal compound begins to dissolve. (One reason for using hydrochloric acid is to convert any metal compound to a chloride — chlorides are more volatile than other salts so are more likely to give better results.)
3. Dip a clean metal wire (platinum or nichrome) or silica rod into the mixture to obtain a sample of the compound.
4. Hold the end of wire or rod in the flame and observe the colour.

SPEC 8.19 ≈ 16 MIN

Tests for Carbonate, Sulfate and Ammonium Ions

THE SPECIFICATION SAYS

know reactions, including ionic equations where appropriate, for identifying: i carbonate ions, CO_3^{2-}, and hydrogencarbonate ions, HCO_3^-, using an aqueous acid to form carbon dioxide (and testing the gas with limewater) ii sulfate ions, SO_4^{2-}, using acidified barium chloride solution iii ammonium ions, NH_4^+, using sodium hydroxide solution and warming to form ammonia (and testing with litmus and HCl fumes)

Tests for ammonium, carbonate and hydrogencarbonate, and sulfate ions — with observations and ionic equations.

Test for ions

ION	TEST	OBSERVATIONS	EQUATIONS
Ammonium NH_4^+	Add aqueous sodium hydroxide and warm	pungent smell (NH_3); damp litmus paper turns blue; or dense white smoke is formed (NH_4Cl) when passed over hydrogen chloride gas	$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$; $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$
Carbonate or hydrogen carbonate CO_3^{2-} or HCO_3^-	Add dilute acid (HCl, H_2SO_4, HNO_3)	Effervescence (CO_2); gas turns limewater ($Ca(OH)_2$) milky ($CaCO_3$)	$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$; $CO_2(g) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + H_2O(l)$
Sulphate SO_4^{2-}	(To a solution of a compound) add aqueous barium chloride followed by dilute hydrochloric acid or nitric acid	White ppt ($BaSO_4$); insoluble in hydrochloric acid	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

Note: there are other ions that could form white precipitate with barium ions, especially carbonate ions, so in the test there must be H^+ ions present to prevent barium carbonate from forming as a white precipitate. Dilute nitric acid or dilute hydrochloric acid is therefore added as part of the test.

SPEC 8.20 ≈ 20 MIN

Solution Concentrations and Titrations

> THE SPECIFICATION SAYS
>
> be able to calculate solution concentrations, in mol dm^{-3} and g dm^{-3}, including simple acid–base titrations using the indicators methyl orange and phenolphthalein

Volumetric analysis: apparatus, standard solutions, concentration calculations, indicators and the mean titre.

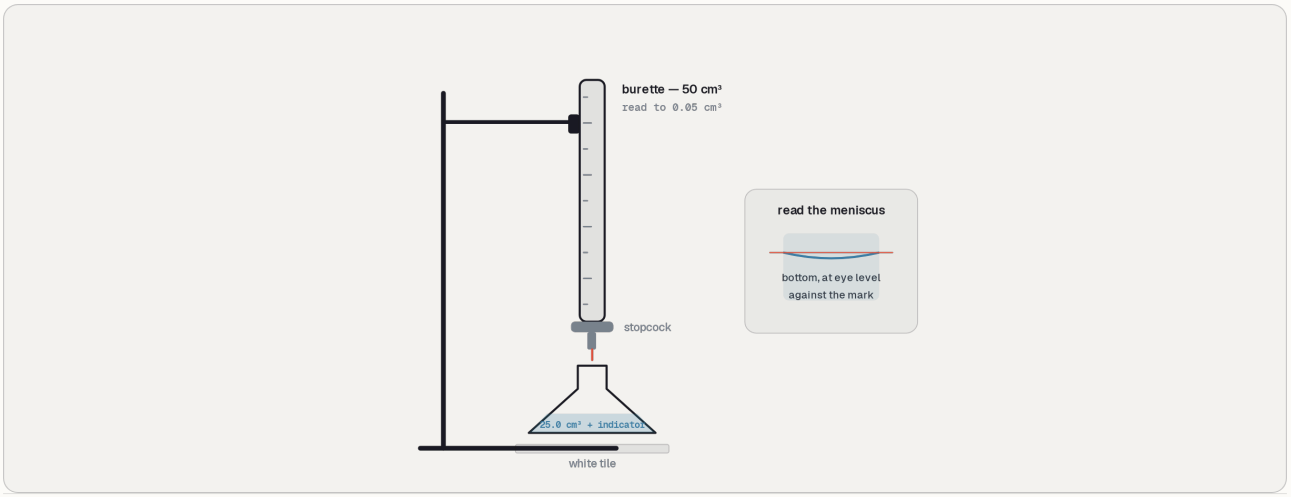
Volumetric analysis

- The quantitative investigation of chemical reactions by comparing reacting volumes is known as **volumetric analysis**.
- The procedure by which reacting volumes are determined is known as a **titration**. One substance needed in a titration is a **standard solution**.
- Volumetric analysis can be used to determine:
- Concentration of a solution
- Formula of a compound
- Number of moles of a substance
- Percentage yield of a substance
- Percentage purity of a substance
- Water of crystallization of a substance

Apparatus used

APPARATUS	USE
Pipette	designed to deliver a fixed volume accurately
Pipette filler	used to fill the pipette
Burette	designed to deliver definite but variable volumes of liquid
Conical flask	used to mix solutions
Volumetric flask	used to prepare standard solutions
Beaker	used to measure solutions approximately
Funnel	used to transfer solutions without spilling
Dropper	used to deliver small volumes of solution
White tile	used to show the colour changes clearly at the end point
Stand	used to attach the burette

SPEC 8.20 · 8.21 · 8.23 · VOLUMETRIC ANALYSIS **The titration bench — burette, pipette, conical flask** n = cV · concordant within 0.20 cm³



THE BENCH Standard solution in one vessel, unknown in the other; end point when the indicator just changes. Mean titre from concordant titres only, then c = n/V — with uncertainties counted (burette read twice).

FIG — *Burette above the conical flask on a white tile; the burette reading is taken from the lowest part of the meniscus.*

Standard solution

- A **standard solution** is a solution whose concentration is **accurately known**. One way to prepare a standard solution is to take a known mass of a substance (primary standards) and dissolve it in water to make a known volume of solution.
- These substances are known as **primary standards**. Ideally primary standards should:
- Be solids with high molar masses
- Be available in a high degree of purity
- Be chemically stable
- Not absorb water from the atmosphere
- Be soluble in water
- React rapidly and completely with other substances when used in titrations

Concentration

- **Concentration** of a solution is usually expressed as the amount of solute dissolved in a given amount of solvent or solution.
- It is usually measured in **moles of solute per dm³ of solution** or in **grams of solute per dm³ of solution**.

Calculations in titration

- Write a balanced chemical equation.
- Calculate the amount/moles of the reagent of known concentration: **amount of substance = concentration × volume (n = cV)**.
- Calculate the number of moles of the unknown substance, using the ratio given in the balanced chemical equation.

- Calculate the concentration of the unknown solution: **concentration = moles/volume (c = n/V)**.

Indicator

- **Indicators** are substances whose solutions change color due to changes in pH (by adding acid or alkali).
- Two common indicators are **methyl orange** and **phenolphthalein**.

INDICATOR	COLOUR IN ACID	COLOUR IN ALKALI	ACID-BASE COMBINATION
methyl orange	red	yellow	strong acid – weak base and strong acid – strong base
phenolphthalein	colourless	pink	weak acid – strong base and strong acid – strong base

Mean titre

- Mean is the set of values added to give a total, which is then divided by the number of values.
- Only **concordant values** must be selected to calculate the mean titre.

TITRATION NUMBER	1	2	3	4
final burette reading / cm³	24.15	25.30	24.60	23.25
initial burette reading / cm³	1.20	2.70	1.90	0.60
titre / cm³	22.95	22.60	22.70	22.65
concordant titres	✗	✓	✓	✓

- Titre 2, 3 and 4 are all within 0.20 cm^3 of each other, so are concordant.

$$\text{Mean} = (22.60 + 22.70 + 22.65) \div 3 = 22.65\ cm^3$$

Only the concordant titres are averaged

SPEC 8.21 ≈ 14 MIN

Core Practical 3: Concentration of Hydrochloric Acid

> THE SPECIFICATION SAYS
>
> CORE PRACTICAL 3 Finding the concentration of a solution of hydrochloric acid.

Core Practical 3: the titration method for finding the concentration of an acid solution.

Titration

- A **titration** is a practical method with the aim of measuring the volumes of two solutions that react together, and using the results to calculate the concentration of one of the solutions.

- It involves having one solution in a **burette** and pipetting a known volume of the other solution (**standard solution**) into a **conical flask**.
- An **indicator** is then added to the solution in the conical flask and is titrated from the burette until the indicator changes colour.

Key vocabulary

- **Equivalence point**: the point at which there are exactly the right amounts of substances to complete the reaction.
- **End point**: the point at which the indicator just changes colour; ideally the end point should coincide with the equivalence point.
- **Meniscus**: the curving of the upper surface of the liquid in a container; the lowest (horizontal) part of the meniscus should be read.
- **Titre**: the volume added from the burette during a titration.
- **Concordant titres**: titres that are close together (usually within 0.20 cm^3 of each other).

Outline of the titration method

- Add the acid to the alkali or vice versa until the equivalence point of the titration and the end point of the indicator is reached.
- Record the burette reading using the lowest part of the meniscus.
- Calculate the titre.
- Repeat the titration until concordant titres are obtained.

Titration method (sulfamic acid and sodium hydroxide)

1. Rinse the conical flask with deionized water and place it on a white tile.
2. Using a pipette filler, rinse the pipette with deionized water and then with some of the sodium hydroxide solution.
3. Use the pipette to transfer 25.0 cm^3 of the sodium hydroxide solution to the conical flask.
4. Add about 3 drops of methyl orange indicator.
5. Rinse the burette with deionized water and then with sulfamic acid solution.
6. Fill the burette with sulfamic acid solution and set it up in the stand above the conical flask.
7. Record the burette reading.
8. Add the sulfamic acid solution to the conical flask until the indicator just changes colour, and again record the burette reading.
9. Empty and rinse the conical flask with deionized water, and repeat the titration until concordant titres have been obtained.

SPEC 8.22 ≈ 16 MIN

Mistakes, Errors and Measurement Uncertainty

> THE SPECIFICATION SAYS
>
> **understand how to minimize the sources of measurement uncertainty in volumetric analysis and estimate the overall uncertainty in the calculated result.**

Mistakes, errors, accuracy and precision, measurement uncertainty and overall percentage uncertainty.

Mistake

- A **mistake** is something that a skilled operator can avoid by being careful.
- Some examples of mistakes:
- A chemist weighs a beaker on a balance without making sure the balance is tared (set to zero) beforehand. The reading on the balance could be very different from the actual mass of the beaker.
- A student sees a burette reading of 27.35 cm^3 but writes it down as 23.75 cm^3. This is the students mistake, and nothing to do with the apparatus.

Error

- **Error** is something that even a skilled operator would find difficult to avoid, and is a consequence of the way the apparatus has been constructed and how readings can be made using it.
- Measurement errors can be divided into **random errors** and **systematic errors**:

TERM	MEANING
Random error	This is an error caused by unpredictable changes in conditions such as temperature or pressure, or by a difference in recording that is difficult to get exactly right.
Systematic error	This is an error caused by the apparatus, and leads to the recorded value being either too low or too high.

Accuracy

- **Accuracy** is a measure of how close values are to the **accepted or correct value**.
- If the pipette is wrongly calibrated, it is not **accurate** even if it is precise. It delivers the same amount of liquid but not the amount it claims to have delivered.

Precision

- **Precision** is a measure of how close values are to **each other**.
- **Burette** is a **precise** device to deliver variable amounts of a liquid measured to the nearest 0.05 cm^3.
- **Pipette** is a **precise** device to deliver a fixed amount of a liquid, usually 25.0 cm^3 or 10.0 cm^3.

Measurement uncertainty

- It is the **potential error** involved when using a piece of apparatus to make a measurement.
- The size of measurement uncertainty is determined by the **precision of the apparatus**.
- Digital balances can have various degree of precision. A balance that measures to three decimal places is more precise than one that measures to only one decimal place.
- The greater the degree of precision of the balance, the smaller the measurement uncertainty in the recorded mass.

NUMBER OF DECIMAL PLACES	MEASUREMENT UNCERTAINTY	EXAMPLE
1	±0.05 g	A reading of 17.1 g could be between 17.05 and 17.15 g
2	±0.005 g	A reading of 17.10 g could be between 17.095 g and 17.105 g
3	±0.0005 g	A reading of 17.100 g could be between 17.0995 g and 17.1005 g

Examples of measurement uncertainties in glassware

APPARATUS	CAPACITY	MEASUREMENT UNCERTAINTY
burette	50 cm^3	±0.05 cm^3 (but the burette is read twice in a titration, so the total measurement uncertainty is ±0.10 cm^3)
pipette	25 cm^3	±0.06 cm^3
volumetric flask	250 cm^3	±0.3 cm^3

Percentage uncertainty

- Percentage uncertainty is the actual measurement uncertainty in an experiment multiplied by 100 and divided by the value recorded.

```
% uncertainty = (actual measurement uncertainty in an experiment ÷ value recorded) × 100
```

Percentage uncertainty

Percentage uncertainty of burette, pipette and volumetric flask

APPARATUS	CAPACITY	UNCERTAINTY	PERCENTAGE UNCERTAINTY
burette	50 cm^3	±0.05 cm^3	Note that two burette values are read in a titration, so the total measurement uncertainty is ±0.10 cm^3. If the titre is 22.50 cm^3, then the percentage uncertainty is: ±0.10 × 100/22.50 = **±0.44%**

APPARATUS	CAPACITY	UNCERTAINTY	PERCENTAGE UNCERTAINTY
pipette	25 cm^3	±0.06 cm^3	The reading is taken only once, and for the same volume each time, so the percentage uncertainty is always: ±0.06 × 100/25 = **±0.24%**
volumetric flask	250 cm^3	±0.3 cm^3	The reading is taken only once, and for the same volume each time, so the percentage uncertainty is always: ±0.3 × 100/250 = **±0.12%**

Percentage uncertainty of balance

- The percentage uncertainty using a balance depends on:
- The precision of the balance
- The mass being weighed

WORKED EXAMPLE **Worked example**

PROBLEM

Mass of marble chips = 3.57 g. The measurement uncertainty in a two decimal place balance is ±0.005 g. We have to count this uncertainty twice — there is a ±0.005 g uncertainty when calibrating the balance to zero, as well as a ±0.005 g uncertainty when measuring mass.

1. Total uncertainty = 2 × 0.005 g.
2. So the percentage uncertainty is ±(2 × 0.005 × 100) ÷ 3.57.

ANSWER **±0.28%**

Total measurement uncertainty

- **Total measurement uncertainty** is obtained by **adding together the individual uncertainties**.
- For example, if a concentration of 0.118 mol dm^{-3} has been calculated from a titration that has involved balance and glassware uncertainties, then the uncertainties might be:

	PERCENTAGE UNCERTAINTY
Balance	±0.09%
Volumetric flask	±0.12%
Pipette	±0.24%
Burette	±0.47%
Overall percentage uncertainty	±0.92%

- This means there is an overall uncertainty in the concentration of 0.92% of 0.118, which is about ±0.001, so the final value can be quoted like this:

$$\text{Concentration} = 0.118 \pm 0.001 \text{ mol dm}^{-3}$$

The exact value is in range 0.117 – 0.119 mol dm^{-3}

Minimizing uncertainty and errors

- Percentage uncertainty can be minimized by using a **higher value** rather than a lower one.
- For example, in using a balance, weighing sample of 5 g will lead to a much lower percentage uncertainty than weighing a 0.5 g sample.
- **Minimizing random errors:** for example, in a thermochemistry experiment, **minimizing heat loss** will reduce random errors in the measurement of temperature change.

SPEC 8.23 ≈ 12 MIN

Core Practical 4: Standard Solution from a Solid Acid

> THE SPECIFICATION SAYS
>
> CORE PRACTICAL 4 Preparation of a standard solution from a solid acid and use it to find the concentration of a solution of sodium hydroxide.

Core Practical 4: preparation of a standard solution from a solid acid.

Preparation of a standard solution

1. **Weigh** accurately the required amount of material (see section 3.2).
2. **Transfer** the material to a beaker and dissolve in a small amount of solvent (usually deionised water). Ensure all the solid has dissolved.
3. Using a clean glass funnel, **transfer the solution quantitatively** into a clean volumetric flask. Wash out the beaker with the solvent a number of times and transfer the washings to the flask. Hint: pouring the liquid down a glass rod held in the spout of the beaker can help prevent liquid running down the side of the beaker.
4. **Make sure the flask and contents are at ambient temperature.** Carefully add solvent to the flask. Use a pasteur pipette to slowly add solvent until the bottom of the meniscus touches the calibration mark on the neck of the volumetric flask (see section 3.1 for information on the correct use of volumetric flasks).
5. **Stopper** the volumetric flask and shake to ensure the solution is thoroughly mixed.

SPEC 8.24 ≈ 16 MIN

Trends in Group 7

> THE SPECIFICATION SAYS
>
> understand reasons for the trends for Group 7 elements in: i melting and boiling temperatures and physical state at room temperature ii electronegativity iii reactivity down the group

The halogens: appearance, changes of state, atomic radius, melting and boiling temperatures, electronegativity and reactivity — with the reason for each trend.

Introduction to the Group 7 elements

- The elements in group VII are known as **halogens** — fluorine, chlorine, bromine, iodine and astatine.
- The term **halogen** is derived from Greek meaning **salt producing**. Halogens are very reactive and readily form salts called **halides**.
- All have the electronic configuration **$ns^2\ np^5$**.
- They exist as separate **diatomic molecules** (F_2, Cl_2, Br_2 and I_2).

Appearance

	F_2	Cl_2	Br_2	I_2
Colour	Pale yellow	Green	Red-brown	Shiny greyish black
State (at RTP)	gas	gas	liquid	solid
In water	very pale yellow	Pale green	Reddish brown	Pale brown
In hydrocarbon solvents	—	Pale green	Reddish brown	Violet

Changes of state

- When bromine is left at room temperature it gives off **brown vapour**: $Br_2(l) \rightarrow Br_2(g)$.
- When iodine is left at room temperature, it gives off brown vapour **without melting**. This change is called **sublimation**: $I_2(s) \rightarrow I_2(g)$.

Atomic radius

- Atomic radius **increases** down the Group 7 as the outer most electrons are further away from the nuclei.

F	Cl	Br	I	
Atomic radius / nm	0.064	0.099	0.111	0.128

Melting and Boiling temperature

- Melting and boiling temperatures **increase** down the group.

- As the size of the halogen increases, the number of electrons also increases.
- This increases the strength of **London forces** (instantaneous dipole-induced dipole attraction).
- Therefore, more energy is required to overcome the forces between molecules.

Electronegativity

DEFINITION

Electronegativity

The ability of an atom to attract the shared pair of electrons in a covalent bond.

- **Electronegativity decreases** down the group. The electronegativity of fluorine is the highest of all elements.
- As the size of the atom increases, the number of shells increases.
- This increases the shielding of the nucleus by the inner shell electrons.
- As a result, the ability of the nucleus to attract electron pair is reduced.

ELEMENT	STATE AT ROOM TEMPERATURE	MELTING TEMPERATURE / °C	BOILING TEMPERATURE / °C	ELECTRONEGATIVITY
Fluorine	gas	−220	−188	4.0
Chlorine	gas	−101	−35	3.0
Bromine	liquid	−7	59	2.8
Iodine	solid	114	184	2.5
Astatine	solid	302	337	2.2

Reactivity

- The reactivity of halogens **decreases down the group**. F_2 is most reactive and I_2 is least reactive.
- The decrease in reactivity can be explained with reference to the same factors used to explain the decrease in electronegativity down the group.
- **Electron affinity** also decreases down the group, therefore it is less easy to form anions.

SPEC 8.25 ≈ 16 MIN

Displacement Reactions and Oxidising Power

THE SPECIFICATION SAYS

understand the trend in reactivity of Group 7 elements in terms of the redox reactions of Cl_2, Br_2 and I_2 with halide ions in aqueous solution. Students are expected to know the colours of the elements in standard conditions, in aqueous solution and in a non-polar organic solvent.

Oxidising power decreases $F_2 > Cl_2 > Br_2 > I_2$ — shown by the displacement reactions with halide ions in aqueous solution.

Displacement reactions of halogen/halide

- Halogens are strong **oxidizing agents** — as they need an electron to complete their outer shell, they accept electrons and get reduced.
- The oxidizing power of the halogen **decreases** as we move down the group from F to I:

$$F_2 > Cl_2 > Br_2 > I_2$$

Oxidising power of the halogens

- As the size of the atoms increases down a group, **shielding increases**.
- Therefore, the ability of the atoms to accept electrons decreases (form −1 ions) and hence their oxidising power decreases.
- **Chlorine** will displace **bromide and iodide** ions from solution.
- **Bromine** will displace **iodide** ions from solution, but not chloride ions.
- **Iodine** cannot displace either bromide or chloride ions from solution.

Worked example

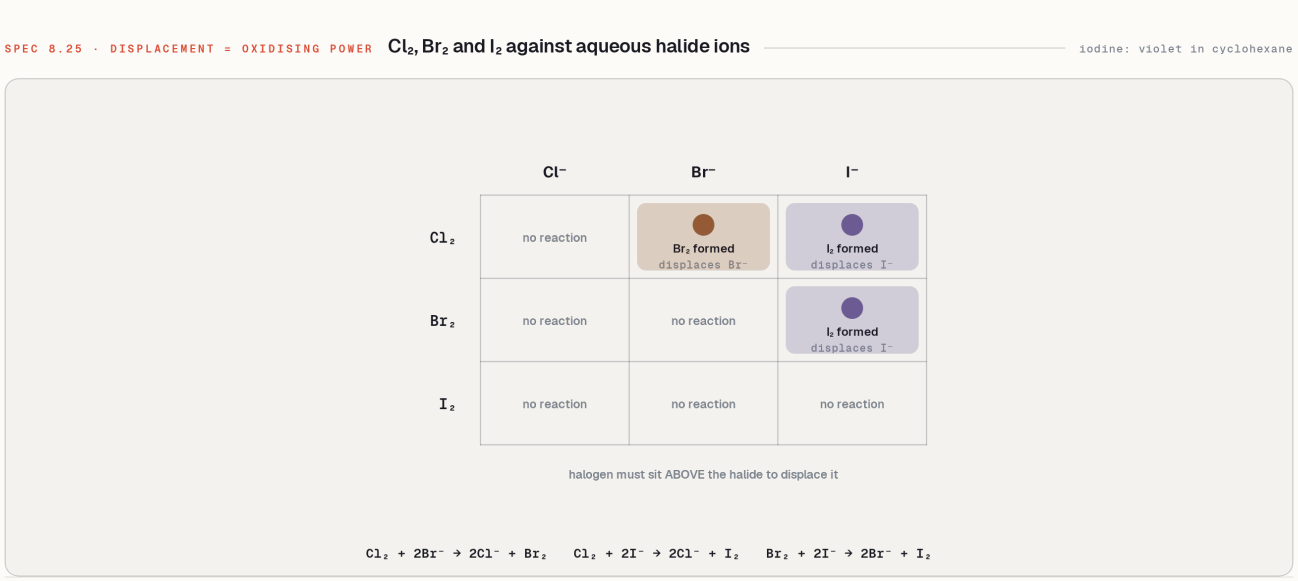
HALOGEN ADDED	OBSERVATION	POTASSIUM CHLORIDE(AQ)	POTASSIUM BROMIDE(AQ)	POTASSIUM IODIDE(AQ)
chlorine(aq)	Appearance of aqueous solution	Very pale green	Reddish brown	Pale brown
chlorine(aq)	Appearance of cyclohexane	Very pale green	Reddish brown	Violet

$$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq) \qquad Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$$

$$Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$$

The ionic equations for the three displacements described above

SPEC 8.25 · DISPLACEMENT = OXIDISING POWER



THE GRID A more reactive halogen oxidises the ions of any halogen below it and is itself reduced (0 → −1) while the halide is oxidised (−1 → 0). Colours are sharpest in the cyclohexane layer.

FIG — *Displacement = oxidising power: each halogen oxidises (and displaces) the halide ions of any halogen below it in the group.*

SPEC 8.26 ≈ 14 MIN

Chlorine with Water and Alkali

THE SPECIFICATION SAYS

understand, in terms of changes in oxidation number, the following reactions of the halogens: i oxidation reactions with Group 1 and 2 metals ii the disproportionation reaction of chlorine with water and the use of chlorine in water treatment. iii the disproportionation reaction of chlorine with cold, dilute aqueous sodium hydroxide to form bleach iv the disproportionation reaction of chlorine with hot alkali v reactions analogous to those specified above

Group 1 and 2 metals with halogens; chlorine with water, with cold dilute alkali and with hot alkali — disproportionation in terms of oxidation numbers.

Metals of Group 1 and 2

- Metals react with halogens on heating to form **halides** or white ionic salts.
- Reactions are most vigorous between elements at the bottom of Group 1 and 2, and elements at the top of Group 7. The most vigorous reaction should be between **caesium (or francium) and fluorine**, and the least vigorous between **beryllium and iodine (or astatine)**.
- These reactions are **redox reactions** in which the halogen acts as an **oxidizing agent**.
- The oxidation number of the halogen **decreases from 0 to −1** and the oxidation number of the metal **increases from 0 to +1 or +2**, depending on the group.

	GROUP I METAL	GROUP II METAL
General equation	$2M + X_2 \rightarrow 2MX$	$M + X_2 \rightarrow MX_2$

$$2Li + Cl_2 \rightarrow 2LiCl \qquad 2Na + Br_2 \rightarrow 2NaBr$$

Group I metal examples

$$Mg + Cl_2 \rightarrow MgCl_2 \qquad Ca + Cl_2 \rightarrow CaCl_2 \qquad Sr + Br_2 \rightarrow SrBr_2$$

Group II metal examples

Disproportionation reactions of chlorine

DEFINITION

Disproportionation

One element undergoes both oxidation and reduction at the same time.

Chlorine with water

- When chlorine is added to water, it dissolves to form **chlorine water**.
- Some of the dissolved chlorine also reacts to form a mixture of two acids, **hydrochloric acid** and **chloric(I) acid** (commonly called **hypochlorous acid**):

$$Cl_2 + H_2O \rightarrow HCl + HClO$$

Cl: 0 → −1 (HCl) · 0 → +1 (HClO)

- **HClO is a mild oxidising agent** and effective at killing bacteria without being harmful to humans. For this reason, a small amount of chlorine dissolved in water will sterilize the water, and chlorine is widely used in **swimming pools** and in **water treatment**.

Chlorine with COLD alkali

- When chlorine is added to **COLD dilute aqueous sodium hydroxide**, it reacts to form the salts of the acids.
- These salts are **sodium chloride** and **sodium chlorate(I)**, which is also known as **sodium hypochlorite**.
- The disproportionation that occurs can be shown using the oxidation numbers:

$$2NaOH(aq) + Cl_2(g) \rightarrow NaCl(aq) + NaOCl(aq) + H_2O(l)$$

Cl: 0 → −1 (NaCl) · 0 → +1 (NaOCl)

- The sodium chlorate(I) formed is also a disinfectant, but it is mainly known for its **bleaching action**. It is used extensively in industry and is the **active ingredient in household bleach**.

Chlorine with HOT alkali

- When chlorine is added to **HOT concentrated sodium hydroxide** solution, it reacts to form sodium chloride and a different product, **sodium chlorate(V)**:

$$6NaOH(aq) + 3Cl_2(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$$

Cl: 0 → −1 (NaCl) · 0 → +5 ($NaClO_3$)

- Chlorine is oxidized to the **+5 state**.
- The sodium chlorate(V) formed is also used in bleaching, and as a **weed killer**.
- **Bromine and iodine react in similar ways.**

SPEC 8.27 ≈ 26 MIN

Reactions of the Halides

THE SPECIFICATION SAYS

understand the following reactions: i solid Group 1 halides with concentrated sulfuric acid, to illustrate the trend in reducing ability of the hydrogen halides ii precipitation reactions of the aqueous anions Cl^-, Br^- and I^- with aqueous silver nitrate solution and nitric acid, and the solubility of the precipitates in aqueous ammonia solution iii hydrogen halides with ammonia gas (to produce ammonium halides) and with water (to produce acids)

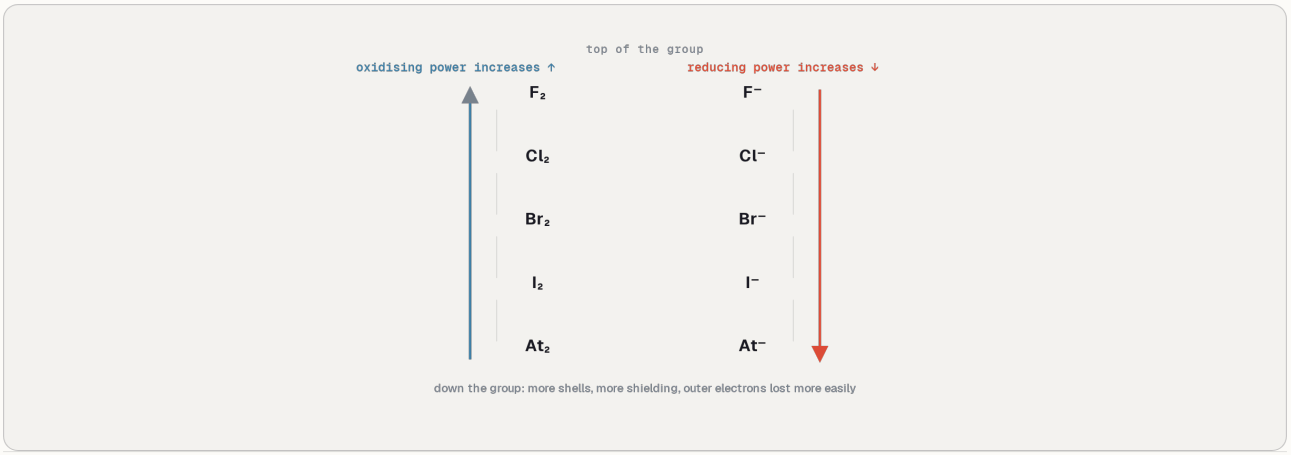
Halide ions as reducing agents — the sodium halides with concentrated sulfuric acid, testing for halide ions with silver nitrate and ammonia, and the hydrogen halides with water and ammonia.

Reducing agent

- The halide ions are **reducing agents**, as they can lose electrons and get oxidized.
- Reducing power of halide ions **increases down the group**. **Iodide ions are the most reducing**, followed by bromide ions. Fluoride ions have **no significant reducing properties**.
- As the number of shells in the ion increases, there is more **shielding** of the nucleus and the outer electrons become less strongly held.
- These electrons are thus **lost more easily** and the halide ion is **more readily oxidised**.

OXIDISING POWER	HALOGEN	HALIDE	REDUCING POWER
High	fluorine F_2	fluoride F^-	Low
↓	chlorine Cl_2	chloride Cl^-	↑
↓	bromine Br_2	bromide Br^-	↑
↓	iodine I_2	iodide I^-	↑
Low	astatine At_2	astatide At^-	High

SPEC 8.25 · 8.27(I) · TWO MIRROR LADDERS **Oxidising power of halogens vs reducing power of halides** oxidising power rises towards F_2 · reducing powe



ONE TREND, TWO SIGNS With concentrated H_2SO_4 the ladder shows itself: Cl^- stops at HCl fumes, Br^- adds $Br_2 + SO_2$, I^- drives the acid down to SO_2, S and even H_2S — while F^- has no significant reducing power at all.

FIG — *One trend, two directions: oxidising power of the halogens rises towards fluorine; reducing power of the halide ions rises towards astatide.*

Reaction of halide ions with concentrated sulfuric acid

- The variation in reducing strength of the halides can be clearly seen in the reaction of the **sodium halides with concentrated sulfuric acid**.
- Concentrated sulfuric acid can act as an **oxidizing agent** and as well as an **acid**.
- When it acts as an oxidizing agent, it is reduced, but the extent of its reduction and the products formed depends on the species being oxidized.
- The three possible reduction products are **sulfur dioxide**, **sulfur** and **hydrogen sulfide**.

NAX	OBSERVATION
NaCl(s)	Misty fumes (HCl)
NaBr(s)	Misty fumes (HBr)
NaBr(s)	Brown fumes Br_2
NaBr(s)	Colourless gas with choking odour (SO_2)
NaI(s)	Misty fumes (HI)
NaI(s)	Purple fumes + black ppt (I_2)
NaI(s)	Colourless gas with choking odour (SO_2)
NaI(s)	Yellow ppt (S)
NaI(s)	Colourless gas with rotten egg smell (H_2S)

Testing for halide ions in solutions

- The analysis of the halide ions is based on the **solubilities of the silver halides**.

- The reagent is **silver nitrate** solution, but dilute **nitric acid** is added first to make sure that any other anions (especially carbonate ions) are removed, as they would also form precipitates.
- If a precipitate is obtained, **dilute or concentrated ammonia** is added to further confirm the halide ion.

HALIDE	ADD DILUTE NITRIC ACID, THEN SILVER NITRATE SOLUTION	TO DISTINGUISH THE PRECIPITATE, ADD DILUTE AMMONIA AND CONCENTRATED AMMONIA SOLUTION
Chloride	White precipitate of AgCl(s)	Precipitate soluble in both
Chloride	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	$AgCl(s) + 2NH_3(aq) \rightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$
Bromide	Cream precipitate of AgBr(s)	Precipitate slightly soluble in dilute ammonia and soluble in concentrated ammonia
Bromide	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	$AgBr(s) + 2NH_3(aq) \rightarrow [Ag(NH_3)_2]^+(aq) + Br^-(aq)$
Iodide	Yellow precipitate of AgI(s)	Precipitate insoluble in both
Iodide	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	—

Hydrogen halides

- Hydrogen halides are **colourless gases** and exist as **polar diatomic molecules**.
- They are formed when hydrogen reacts with halogens.

Reactions with water

- The hydrogen halides readily react with water to form **acidic solutions**, all of which are colourless.

HYDROGEN HALIDE	ACID FORMED	EQUATION
Hydrogen fluoride	hydrofluoric acid	$HF + H_2O \rightleftharpoons H_3O^+ + F^-$
Hydrogen chloride	hydrochloric acid	$HCl + H_2O \rightarrow H_3O^+ + Cl^-$
Hydrogen bromide	hydrobromic acid	$HBr + H_2O \rightarrow H_3O^+ + Br^-$
Hydrogen iodide	hydroiodic acid	$HI + H_2O \rightarrow H_3O^+ + I^-$

Reaction with ammonia

- Hydrogen halides react with ammonia gas to form salts, all of which are **white ionic solids**:

$$HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$$

Example — ammonium chloride, a white ionic solid

SPEC 8.28 ≈ 8 MIN

Predicting Fluorine and Astatine

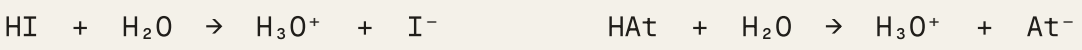
> THE SPECIFICATION SAYS
>
> **be able to make predictions about fluorine and astatine and their compounds, in terms of knowledge of trends in halogen chemistry**

Predicting reactions of fluorides and astatides using the trends in halogen chemistry.

If you are asked to predict reactions of **fluorides and astatides** that you are not familiar with, you can use information in this topic to help you.

Fluorides and Astatides

- For example, you could write an equation to show the formation of an acid when **hydrogen astatide** is added to water, based on the equation for hydrogen iodide:



The hydrogen astatide prediction, based on the equation for hydrogen iodide

Where fluorine and astatine already appear in this topic

- **Appearance and physical state:** fluorine is a pale yellow gas; astatine is a solid at room temperature.
- **Melting and boiling temperatures:** astatine continues the increase down the group — melting temperature 302 °C, boiling temperature 337 °C.
- **Electronegativity:** fluorine is the highest of all elements (4.0); astatine continues the decrease (2.2).
- **Reactivity and oxidising power:** fluorine is the most reactive halogen, at the top of the order $F_2 > Cl_2 > Br_2 > I_2$; astatine sits at the low end of the oxidising-power ladder, so astatide At^- sits at the high end of the reducing-power ladder.
- **Reactions with metals:** the least vigorous reaction of all is between beryllium and iodine (or astatine).

END OF TOPIC 08

01 · Formulae, Equations and Amount of Substance | 02 · Atomic Structure and the Periodic Table | 03 · Bonding and Structure | 04 · Introductory Organic Chemistry and Alkanes | 05 · Alkenes | 06 · Energetics | 07 · Intermolecular Forces | 09 · Introduction to Kinetics and Equilibria | 10 · Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

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